EXP07

CODE:

```
import torchfrom transformers import VisionEncoderDecoderModel,
ViTImageProcessor, AutoTokenizerfrom PIL import Imagefrom
google.colab import filesfrom IPython.display import display

# Step 3: Load pretrained image captioning modelmodel_name =
"nlpconnect/vit-gpt2-image-captioning"
print("Loading model...")
model = VisionEncoderDecoderModel.from_pretrained(model_name)
processor = ViTImageProcessor.from_pretrained(model_name)
tokenizer = AutoTokenizer.from_pretrained(model_name)

# Step 4: Select devicedevice = torch.device(
    "cuda" if torch.cuda.is_available() else "cpu")
model.to(device)model.eval()
print("Model loaded successfully!")print("Using device:", device)

# Step 5: Define caption generation functiondef
generate_caption(image_path, max_length=30, num_beams=4):

    # Open image
    image = Image.open(image_path).convert("RGB")

    # Convert image into model input
    pixel_values = processor(
        images=image,
        return_tensors="pt"
    ).pixel_values.to(device)

    # Generate caption
    output_ids = model.generate(
        pixel_values,
        max_length=max_length,
        num_beams=num_beams,
        early_stopping=True
    )

    # Convert output tokens into tex
    caption = tokenizer.decode(
        output_ids[0],
        skip_special_tokens=True
    )
```

```
    return caption

# Step 6: Upload imageprint("\nPlease upload an image:")
uploaded = files.upload()
image_path = list(uploaded.keys())[0]
print("Image uploaded:", image_path)

# Step 7: Generate captioncaption = generate_caption(image_path)

# Step 8: Display imageprint("\nInput
Image:")display(Image.open(image_path))

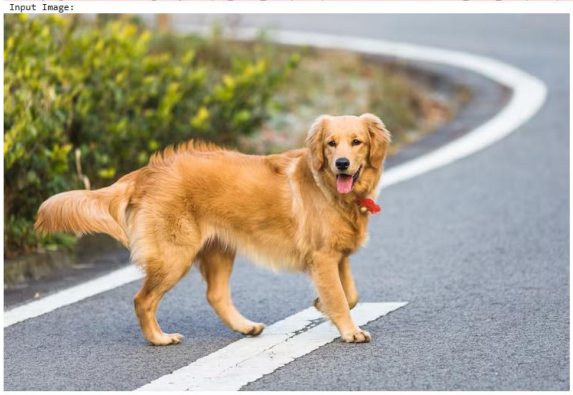
# Step 9: Display generated captionprint("\nGenerated
Caption:")print(caption)
```

OTUPUT:

```
Loading model...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
[transformers] VisionEncoderDecoderModel LOAD REPORT from: nlpconnect/vit-gpt2-image-captioning
Key                                                  | Status     |  |
-----------------------------------------------------+------------+--+-
decoder.transformer.h.{0...11}.attn.masked_bias      | UNEXPECTED |  |
decoder.transformer.h.{0...11}.crossattention.masked_bias | UNEXPECTED |  |

Notes:
- UNEXPECTED:    can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Model loaded successfully!
Using device: cpu

Please upload an image:
Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving happy-pet-dogs-playing-park_1359-281.avif to happy-pet-dogs-playing-park_1359-281.avif
Image uploaded: happy-pet-dogs-playing-park_1359-281.avif
[transformers] We strongly recommend passing in an `attention_mask` since your input_ids may be padded. See https://huggingface.co/docs/transformers/troubleshooting#incorrect-output-when-padding-tokens-arent-masked.
You may ignore this warning if your `pad_token_id` (50256) is identical to the `bos_token_id` (50256), `eos_token_id` (50256), or the `sep_token_id` (None), and your input is not padded.
Input Image:
```



```
Generated Caption:
a brown dog standing on the side of a road
```